SFU CubeSAT: Internship Program

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As part of a six-week summer research internship with the CubeSAT team of Saint Francis University, I was privileged enough to combine my class education with practical aerospace engineering, software development, and system integration. Setting specific goals for myself that would involve improving my programming skills in Python, understanding Git as a version-control tool in team environment, and enhancing my technical problem-solving skills, I was immersed in the process of constructing and testing a working FlatSat. This paper describes my experience in achieving those educational objectives within the internship and outlines some of the most important outcomes of the program based on the book written by Gene Kim called *"The Phoenix Project."*

How Learning Outcomes Have Been Met

Before beginning my six-week internship with the Saint Francis University CubeSAT team I determined three learning objectives that I aimed to complete during this time. To begin with, I wanted to better my programming skills in the Python programming language. Second, I wanted to experience programming in a team and overall gain a better understanding of using Git. The last objective I set was to better my troubleshooting skills. Overall, I met all these objectives.

The first objective, to better develop my programming skills in Python, was met during the six-week internship in multiple diverse ways. During the first week, I developed a Python-based program to ensure the Raspberry Pi 4 is healthy before running the Mode Manager script. This program will also be used in the future to ensure the Raspberry Pi 4 remains healthy after

completion of radiation testing. Then, throughout the third week, I refactored the teams existing Mode Manager program to better simulate how the program will work while the satellite is deployed in space. I updated the codebase from a single 1000+ line file, into multiple 100-line files where each file contains a single class for each operational mode. Throughout the remaining three weeks, I continued updating the "new" Mode Manager program whenever I encountered an issue or area that needed updating. With the completion of a Python-based health monitoring script, and the completion of writing and updating the Mode Manager script, I would say I improved my programming skills in Python.

The second objective that I completed throughout my internship was to gain experience in programming as a team and have more experience with using Git. Git is a version control software that the software team is using to track changes, and work on the software without affecting what other developers are working on. Since I was not the only developer working on this project and will not be only developer working on this project in the future it is vital to have an advanced understanding of Git. Throughout this internship I had to clone our GitHub repository as well as became very familiar with making commits, pushing, pulling, making branches, and merging branches, therefore I believe that I met the second objective of programming in a team environment at gaining a better understanding of Git.

The last objective that I set for myself was to better my troubleshooting skills. Out of the three objectives I set for myself, this is the one I made the most progress with. Throughout the six-week program, I encountered several issues with the physical components, like the cameras, x-ray sensor, and HF receiver, and had to troubleshoot how they communicated with the software. In the end, I was able to develop a process for troubleshooting, starting with finding what works and introducing a new unfamiliar element one at a time. In conclusion, I believe all

three learning objectives I set before the beginning of the six-week Saint Francis University CubeSAT internship program.

CORE 15 Student Learning Outcomes

As defined in the syllabus, there are four CORE 15 student learning outcomes that must be met. I believe that all four have met. The first learning objective states, “Respond to situations in work and life with innovation, flexibility, and problem-solving.” This objective was met through my experience with troubleshooting, often situations would arise where I would need to prioritize different things and think on my feet. The second objective from the syllabus says, “Demonstrate an appreciation of Franciscan values, Catholic Social Teaching, and other religious perspectives to enhance oneself and enrich the community for the common good of all.” Throughout this internship I had the ability to help others in many ways. One way is through work, if I had extra time on my hands, I would use that time to help others with what they were working on, often this meant soldering or organizing different things in the workspace. The third objective states, “Analyze and navigate economic, social, and political systems essential to building and strengthening community.” This outcome was met because navigating economic systems was necessary. For example, I had to speak with vendors from all over the world from places like South Africa and Ecuador regarding various products and services. The final objective is, “Become an informed, open-minded citizen who respects the importance of inclusion and accepts the inherent value and individuality of every person.” This final objective was met because a CubSAT project brings together a group of people from diverse backgrounds, so this objective was met. In conclusion, the CORE 15 student learning outcomes, defined in the syllabus, were met throughout the six-week internship program.

Self-Evaluation

During the Saint Francis University CubeSAT Internship Program, I learned a lot about CubeSATs in general, as well as collaborating with others to meet a common goal. As a goal, set by an advisor, by the end of the six weeks we were to be able to demonstrate using our FlatSat. The FlatSat is a way of testing all our physical components, like the two cameras, x-ray sensor, and HF receiver, along with our software in an environment where everything is laid out on a table. This is so that if we find any issues throughout testing it is easy to isolate the issue, instead of taking apart a testing satellite every time we find an issue. Throughout the program I faced troubleshooting issues, product delivery delays, physical component issues, and software issues, but in the end my group was able to quickly resolve issues and successfully build and test our FlatSat. Overall, overall this was a positive experience for me and an overall success for the CubeSAT program.

Review of The Pheonix Project by Gene Kim

The novel *"The Phoenix Project"* by Gene Kim explains the impact that systematic constraints, unstructured work, and lack of proper integration may have on a large and complex project. This concept is clear from the numerous issues that have arisen in our FlatSat project development process. In a similar fashion to the protagonist of the novel who is trying to deal with chaos at his place of employment, we encountered various technological constraints when working on our project. From handling repository overload by dealing with the problem of 100

MB logs to resolving the problem of power delivery problems of the 630 nm camera, among other physical components, root cause analysis was required everywhere.

Moreover, our practical experience proved in its entirety the basic tenets of DevOps and the "Three Ways" that Gene Kim stressed. We translated our quick feedback mechanisms into practical steps, including repairing the loose solder connections, altering the pin connectors, and creating separate power distribution cables to eliminate hardware problems. We successfully verified our baseline payload separately and simplified the data collection process through a step-by-step sequence during capstone alignment to reach a consistent and fully integrated system. In the end, the internship became a notable example of The Phoenix Project in real life.

Conclusion

In conclusion, the Saint Francis University CubeSAT internship program was a defining experience in terms of a capstone project that effectively fulfilled all my individual learning goals, as well as the general CORE 15 student learning outcomes. Every single challenge from dealing with complex python code bases and collaborative processes using Git to solving hardware constraints and power delivery problems and crosstalk sensors became a lesson in structured problem solving. Through correlating our day-to-day laboratory experiments such as camera power bus isolation and switchboard troubleshooting to the project management concepts introduced in *"The Phoenix Project,"* we got a better understanding of the importance of cross-functionality and structured integration.